

Machine Learning Fundamentals

1. Machine Learning

Machine learning is a field of artificial intelligence in which systems learn patterns from data and use those patterns to make predictions or decisions.

A typical machine learning workflow includes collecting data, cleaning data, selecting features, splitting the dataset, training a model, evaluating performance, and deploying the model.

2. Supervised Learning

Supervised learning uses labeled examples where the input and expected output are known. The model learns a relationship between input features and target values.

Regression predicts continuous values, such as house prices or temperature. Classification predicts categories, such as spam or not spam.

3. Unsupervised Learning

Unsupervised learning works with data that does not have labeled target values. The goal is to discover hidden structures or patterns.

Clustering algorithms group similar data points. K-Means is a common clustering algorithm used for customer segmentation and exploratory analysis.

4. Model Evaluation

Classification models can be evaluated using accuracy, precision, recall, and F1-score. A confusion matrix helps visualize correct and incorrect predictions.

Regression models commonly use Mean Absolute Error, Mean Squared Error, and Root Mean Squared Error to measure prediction error.

5. Overfitting and Underfitting

Overfitting occurs when a model learns the training data too closely and performs poorly on unseen data. Underfitting occurs when the model is too simple to capture important patterns.

Techniques such as cross-validation, regularization, feature selection, and suitable model complexity can help improve generalization.